Lesion-Anchored Prototypes: Reducing Artifact Shortcuts in Interpretable Melanoma Classification

Yousef Azizi Movahed
Dept. of Computer Science, School of Mathematics, Statistics, and Computer Science, College of Science
University of Tehran, Tehran, Iran
azizi.movahed@ut.ac.ir

Fatemeh Ziaeetabar*
Dept. of Computer Science, School of Mathematics, Statistics, and Computer Science, College of Science
University of Tehran, Tehran, Iran
fziaeetabar@ut.ac.ir

***Abstract*—Melanoma classifiers pick up artifacts such as dark corners, rulers and ink. In a prototype network, the explanation itself ("this part looks like that prototype") can point at the artifact. On ISIC 2019, dark corners and bright structures are associated with melanoma even after stratifying by source (Mantel–Haenszel odds ratios 1.555 and 1.751). Existing remedies supervise accuracy or activations; none controls *which* training patches a prototype may become. We propose lesion-anchored prototype learning with two components: a lesion-attraction objective on the normalized spatial attention of each prototype, and a projection step that admits only artifact-free lesion patches, so that every prototype is lesion tissue by construction. The masks need no new annotation and are validated (Dice 0.94 on held-out expert masks, 0.92 on PH2); inference needs none. With identical backbone, leakage-audited split and budget, it raises the test AUC of ProtoPNet from 0.825 to 0.865, outperforms the IAIA-BL mask loss on the same masks (0.844; every seed of ours beats every prototype-baseline seed), places all prototypes on the lesion, and lowers the artifact share of the explanation from 6.6% to 4.8%. When artifacts are inserted or removed around bit-identical lesion pixels, it roughly halves the decision-flip rate of ProtoPNet for dark corners (10.9%→5.2%) and rulers and keeps its explanations stable (Spearman 0.92 vs. 0.68). On unedited images, dark corners double the false-positive rate of the mask-loss model on benign lesions (16%→33%) but barely change ours (17%→20%). An extra artifact-suppression term brought no significant gain (a negative result). Code and logs: https://github.com/YousefAMovahed/lesion-anchored-prototypes.**

***Index Terms*—explainable AI, prototypical part networks, shortcut learning, dermoscopy artifacts, melanoma, counterfactual evaluation**

I. Introduction

Convolutional networks can classify skin lesions as accurately as dermatologists [1], but they also learn *shortcuts* [2], i.e. features that happen to correlate with the label in the training data without being related to the disease. Dermoscopy offers several documented cases. Surgical skin markings raise the melanoma probability of a deep classifier [3], and models still perform far above chance when the lesion is removed from the image [4], [5]. Dark corners, rulers, gel bubbles and ink are frequent in public archives and unevenly distributed across acquisition sites [6]–[8].

For clinical use, Rudin [9] argues for models that are interpretable by design rather than explained after the fact. The prototypical part network (ProtoPNet) [10] compares image patches with learned prototypes, each projected onto a real training patch, so that a decision reads "this part looks like that part". ProtoPNet does not, however, stop a prototype from being a piece of a ruler or of a black corner. In that case the shortcut survives and is in addition presented to the user as a reasonable explanation [11]; in dermoscopy, such prototypes have been observed on black borders, image corners and colored patches [12], [13]. **Research gap.** Prior work leaves three things open (Table I). (i) Debiasing and bias-correction methods for dermoscopy [7], [14], [15] improve robustness to artifacts, but their models are black boxes, so a user cannot see whether a decision rests on the lesion. (ii) For prototype networks, mask supervision penalizes activation outside annotated regions [16], also with automatically segmented lesions [12], and human feedback removes or switches off unwanted prototypes [13], [17]. None of these methods constrains the *push* step, which decides which training patch a prototype finally *is*, so a prototype can still be projected onto a ruler or a black corner, and feedback-based methods act only after a user has found it. (iii) Artifact insertion has been used to test the *decisions* of classifiers [13], [15], [18], but whether the *explanations* of a prototype network stay on the lesion when artifacts are added or removed has, to our knowledge, not been tested; accuracy and qualitative heatmaps remain the usual evidence.

Contributions. This paper fills these gaps with the following contributions.

- **Lesion attraction on the normalized prototype attention.** A lesion-attraction objective acts on the spatial attention of every prototype and pulls it towards lesion tissue without shrinking the activations. It needs no new annotation and drives the accuracy gain: removing it lowers the test AUC from 0.865 to 0.830 ($p = 8 \times 10^{-8}$), and the full model gains 4.0 AUC points over ProtoPNet and 2.1 over the IAIA-BL mask loss applied to the same masks, the setting of the closest prior work [12]. It also suppresses the real dark-corner shortcut that this mask loss leaves in place.
- **Lesion- and artifact-constrained projection.** We restrict the push step to artifact-free lesion patches. Every pro-

*Corresponding author: Fatemeh Ziaeetabar (fziaeetabar@ut.ac.ir).

totype is then lesion tissue *by construction*, a guarantee that loss-based supervision does not give: with the mask loss, only 10 of 20 prototypes land on the lesion and 2 on artifacts. The ablation shows that this step leaves the accuracy unchanged but halves the share of the explanation captured by inserted dark corners and rulers.

- **A counterfactual test of decisions *and* explanations.** Beyond the decision-level insertion tests of earlier work, we measure how far a prototype explanation moves. Four artifact types are inserted, and real hair and ruler marks are removed, while every lesion pixel stays bit-identical; an inpainting-only null control separates the effect of the artifact from that of the editing, and a test on unedited images complements the synthetic edits.
- **A leakage-controlled and mask-audited evaluation.** We group near-duplicates in addition to lesion identifiers, document and validate the provenance of every mask, show that the classifier needs no mask at test time, compare eight configurations with the same backbone over three seeds, and test on three external sets.

An artifact-suppression loss that we also explored did not improve accuracy or counterfactual robustness significantly; we keep it in the ablation and discuss it as a largely negative result rather than as a contribution.

II. Related Work

Artifacts and shortcuts in dermoscopy. Bissoto *et al.* showed that ISIC models perform well even when the lesion is occluded [4], annotated seven artifact types, found that simple debiasing does not remove the bias [5], and later used artifact annotations to build domain-generalization benchmarks [6]. Winkler *et al.* [3] reported that skin markings raise the melanoma score of a commercial-grade classifier, and Pewton and Yap [8] labeled dark corners in ISIC images and found that removing them changes accuracy only marginally but moves network activations onto the lesion. Nauta *et al.* [18] showed that pasting colored calibration patches into images of malignant lesions makes a CNN call them benign. EPVT [7] treats artifact groups as domains and learns domain prompts for a vision transformer, Bevan and Atapour-Abarghouei [14] remove artifact information from the representation by adversarial unlearning, and Reveal-to-Revise (R2R) [15] finds band-aids, rulers and skin markers in ISIC 2019 semi-automatically, corrects the model with RRR, CDEP or ClArC, and evaluates it on test images with pasted artifacts. All of these models are black boxes.

Prototype-based interpretability. ProtoPNet [10] classifies by similarity to class-specific prototypes that are projected onto real training patches, which makes its reasoning inspectable [9]. Hoffmann *et al.* [11] showed that the visual similarity suggested by such prototypes can be misleading, which motivates testing explanations rather than taking them at face value [19].

Explanation supervision and prototype debugging. Right-for-the-right-reasons (RRR) [20] penalizes input gradients inside annotated irrelevant regions, and CDEP [21] penalizes contextual-decomposition importance, demonstrated on ISIC images with colored patches. IAIA-BL [16] adds a fine-annotation loss to a prototype network that penalizes prototype activation outside regions marked by radiologists. Closest to us, Correia *et al.* [12] apply this loss to melanoma-versus-nevus classification with lesion masks from a segmentation network; our PPN-FA baseline follows this approach. ProtoPDebug [17] and the medical study of PIP-Net [13] let a user mark confounded prototypes, which are then forgotten or switched off. None of these methods constrains the projection step, which decides what a prototype finally is, and none tests prototype explanations under artifact interventions. Concurrently with our work, Lin and Wright [22] restricted the projection to patches with sufficient foreground overlap on natural images; we add an artifact-exclusion condition and quantify the effect of the constraint on shortcut artifacts by ablation and counterfactual tests.

TABLE I

Positioning against related work. Annot.: human input required; Interp.: interpretable by design; Where: constrains where the model looks; Prov.: constrains which training patches prototypes become; CF: artifact interventions on test images used to test decisions (D) and explanations (E). Lin and Wright [22] study natural images.

Method	Annot.	Interp.	Where	Prov.	CF
EPVT [7], [14]	image labels	–	–	–	–
R2R [15]	semi-auto.	–	✓	–	D, E
RRR [20], CDEP [21]	pixel masks	–	✓	–	–
ProtoPNet [10]	none	✓	–	–	–
IAIA-BL [16]	expert masks	✓	✓	–	–
Correia *et al.* [12]	auto. masks	✓	✓	–	–
ProtoPDebug [17]	user	✓	✓	post hoc	–
PIP-Net (med.) [13]	user	✓	–	post hoc	D
Lin, Wright [22]	object masks	✓	✓	foreground	–
Ours	**none new**	✓	✓	**les., no art.**	**D, E**

III. Method

A. Prototypical part network

We build on a DenseNet-121 backbone [23]. Two 1×1 convolutions with a final sigmoid turn its output into a latent map $z \in [0,1]^{H \times W \times D}$, here with $H = W = 7$ and $D = 128$. The network keeps $m = 20$ prototypes $p_j \in \mathbb{R}^D$, ten for each class, and $c(j)$ denotes the class of prototype j. For every cell we compute the distance $d_j(h,w) = \|z_{hw} - p_j\|_2^2$ and convert it into a similarity $s_j(h,w) = \log \frac{d_j(h,w)+1}{d_j(h,w)+\epsilon}$ with $\epsilon = 10^{-4}$. A prototype is as active as its best-matching cell, $a_j = \max_{h,w} s_j(h,w)$, and the logits are $\mathbf{W}\boldsymbol{a}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_c \mathcal{L}_{\mathrm{clst}} - \lambda_s \mathcal{L}_{\mathrm{sep}} + \lambda_l \mathcal{L}_{\mathrm{les}} + \lambda_a \mathcal{L}_{\mathrm{art}}, \quad (1)$$

in which $\mathcal{L}_{\mathrm{CE}}$ is a class-weighted cross-entropy, while the cluster and separation costs measure the distance to the closest prototype of the own and of the other class [10]. We keep the original weights $\lambda_c = 0.8$ and $\lambda_s = 0.08$. The proposed approach consists of the lesion-attraction term $\mathcal{L}_{\mathrm{les}}$ ($\lambda_l = 1$) and the constrained projection of Section III-C; the artifact-suppression term $\mathcal{L}_{\mathrm{art}}$ ($\lambda_a = 1$ in our main runs, $\lambda_a = 0$

in the ablation) is examined separately in Section VI-G. All three use lesion and artifact masks whose construction and validation are described in Section IV.

B. Lesion attraction (core component)

During training the masks undergo the same geometric augmentation as the image. Average pooling to the 7×7 grid then gives, for each cell, the fraction covered by lesion, $M^L(h, w)$, and by artifacts, $M^A(h, w)$. We describe where a prototype looks by the spatial attention $\alpha_j = \mathrm{softmax}_{h,w}(s_j/\tau)$, $\tau = 0.5$, a soft version of the max-pooling location that produces the activation a_j. Each prototype is weighted by $\omega_j = a_j / \sum_k a_k$ (treated as a constant), so that the prototypes the model actually uses count most. With $L_{ij} = \sum_{h,w} \alpha_{ij} M_i^L$, the attention mass that prototype j places on the lesion of image i,

$$\mathcal{L}_{\mathrm{les}} = -\frac{1}{|\mathcal{B}_L|} \sum_{i \in \mathcal{B}_L} \sum_{j:\, c(j)=y_i} \tilde{\omega}_{ij} \log\left(L_{ij} + \delta\right), \qquad (2)$$

where $\tilde{\omega}$ renormalizes ω over the prototypes of the true class y_i, $\delta = 10^{-4}$, and $\mathcal{B}_L$ holds the images of the batch whose crop still contains lesion. The logarithm penalizes prototypes that ignore the lesion almost completely much more strongly than those that already look at it. Because α_j sums to one, raising the lesion mass necessarily lowers the mass on everything else, artifacts included. Unlike the fine-annotation loss of IAIA-BL [12], [16], which penalizes the magnitude of activation outside a mask, $\mathcal{L}_{\mathrm{les}}$ acts on the normalized attention, so it does not push activations down globally and leaves their scale to the classification losses.

C. Lesion- and artifact-constrained projection (core component)

In the "push" step, ProtoPNet replaces each prototype with the closest latent patch from a training image of the same class. Without a constraint, nothing prevents this patch from lying on a ruler, a black corner or healthy skin, even if the attention of the prototype was well placed during training. We therefore only admit cells with $M^L \geq 0.5$ and $M^A \leq 0.05$ as candidates. Every prototype is then, by construction, a lesion patch without detected artifacts, and the "that part" of each explanation is guaranteed to be lesion tissue. After the push we train $\mathbf{W}$ alone and keep the backbone, the add-on layers and the batch-normalization statistics fixed, which keeps each prototype identical to the patch it came from. In our runs, five last-layer epochs were not enough to solve this convex problem for *any* of the prototype models. We therefore re-optimize $\mathbf{W}$ to convergence on fixed training activations (weighted cross-entropy plus an ℓ_1 penalty of 10^{-4} on negative-class connections [10]; 3,000 full-batch Adam steps with a cosine-annealed learning rate starting at 0.02). We apply this step in the same way to every prototype model, without touching the validation data, and also report results without it.

IV. Masks: Provenance, Reliability and Leakage Control

Both core components depend on lesion masks. We describe where every mask comes from, how reliable it is, how it is used and why it cannot leak test information.

A. Provenance of the lesion masks

Two sources are used, and every image is tagged with its source. (i) *Expert masks.* For the 10,015 HAM10000 images (39.5% of ISIC 2019) we use the publicly released lesion segmentations of HAM10000 [24] without modification. (ii) *Predicted masks.* The 15,316 BCN20000 [25] and MSK [26] images have no public masks. For them we train a DeepLabV3 [27] network with an ImageNet-initialized MobileNetV3-Large [28] encoder on 256×256 images (12 epochs, AdamW, one-cycle learning rate up to 5×10^{-4}, binary cross-entropy plus soft Dice, random flips and 90° rotations). Its training set consists *only* of the 8,163 HAM10000 images of our training split; the split is fixed before the segmentation network is trained. At inference the prediction is averaged with that of the horizontally flipped image, thresholded at 0.5, reduced to its largest connected component and hole-filled. A mask is marked invalid if it covers less than 0.5% of the image (19 images, 0.08%); invalid masks are excluded from the mask-based losses and metrics.

B. Validation of the lesion masks

On the 1,852 HAM10000 images of the validation and test splits, which the segmentation network never saw, the predicted masks reach a mean Dice score of 0.9445 (median 0.9672, mean Jaccard 0.9015) against the expert masks. To check generalization to another clinic and device, we also evaluated the network, without any adaptation, on the 200 expert-segmented images of PH2 [29] from Portugal: the Dice score is 0.918. We also measure directly whether our conclusions depend on mask quality: Section VI-H repeats the analysis separately on test images with expert and with predicted masks, and with lesion masks deliberately eroded or dilated by 5 pixels.

C. Artifact masks

Artifact masks are produced by fixed, label-free image operations at 256×256 pixels; they contain no learned parameters and always exclude the dilated lesion. *Dark corners* (DC) are dark regions (HSV value < 50) that touch the image border and are at least 8 px thick on average. *Thin dark structures* (TS), which cover hair and ruler ticks, are found with a 13×13 black-hat filter, as in DullRazor [30], followed by an elongation test. *Bright structures* (SP), i.e. specular highlights and the rims of gel bubbles, come from an 11×11 white top-hat filter, and *ink or colored patches* (INK) from strongly saturated blue-to-violet pixels. We flag an image for a type once the mask covers more than 1%, 0.3%, 0.05% or 0.1% of it, respectively. We inspected 24 randomly chosen flagged images for each of the two most frequent types: all 24 DC flags were correct, and 20 of the 24 TS flags marked hair or a

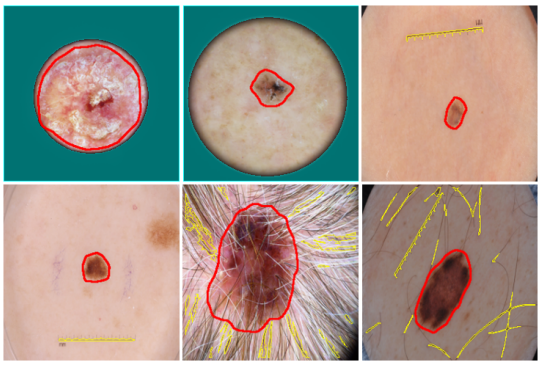


Fig. 1. Automatically generated masks on ISIC 2019 images (red: lesion). Top left and center: dark corners (shaded cyan); top right and bottom left: rulers and bottom center/right: hair, detected as thin dark structures (yellow). Artifact masks never overlap the lesion.

ruler, while the other four were triggered by vessels, pigment structures or skin texture. The SP detector also fires on light hairs, so its masks should be read as "bright structures" rather than strictly as gel bubbles. Examples are shown in Fig. 1.

D. How masks are used, and why they cannot leak

Masks enter the pipeline at exactly four places: the training losses and the constrained projection, and, in the evaluation, the attribution metrics and the placement of counterfactual edits. The classifier itself never receives a mask: at test time it sees only the image, so the classification results do not depend on whether a test mask is correct. Leakage is prevented at every stage.

- *Split first.* The train/validation/test split is computed from image identifiers, lesion identifiers and image content only, before any model, including the segmentation network, is trained.
- *Group-level split.* Images are grouped by lesion identifier and, in addition, by near-duplicate pairs, so that detected re-acquisitions and resized copies of the same lesion cannot end up in different sets (Section V-A). An audit of the final split finds no shared group, no shared lesion identifier and no shared image between any two of the three sets, and all 342 near-identical pairs (thumbnail correlation ≥ 0.97) as well as all other verified near-duplicate pairs lie within a single set.
- *Training-only supervision.* The segmentation network is trained on the 8,163 training-split masks only; the 1,852 expert masks of the validation and test splits never enter any training and serve only for evaluation (segmentation accuracy, attribution metrics and counterfactuals). The mask-based losses and the projection use training images only. Thresholds and checkpoints are chosen on the validation split, and the last-layer re-optimization uses training activations only.
- *Label-free masks.* Neither the segmentation network nor the artifact detectors see a diagnosis, so a mask cannot carry label information from one image to another.
- *Untouched external sets.* Derm7pt, PAD-UFES-20 and PH2 are used only for evaluation. External images flagged as possible near-duplicates of ISIC training images (8, 6 and 7 images) change no external AUC by more than 0.004 when they are removed.

V. Experimental Setup

A. Data and leakage-free split

The task is melanoma versus all other diagnoses on the 25,331 training images of ISIC 2019 (HAM10000 [24], BCN20000 [25] and MSK [26]), of which 4,522 (17.9%) are melanomas. All images are resized to 256×256. Unlike common practice, we keep the 2,084 MSK images that lack a lesion identifier. To avoid leakage we group images not only by lesion identifier but also by near-duplicate pairs (perceptual-hash distance ≤ 8 and a thumbnail correlation of at least 0.9 for both the full and the central image). Grouping look-alike pairs errs on the safe side, since it can only remove leakage. The 13,259 resulting groups are split into ten stratified folds, one for testing and one for validation; each source is split about 81/10/9%. Table II shows that the diagnoses, the three sources, the two kinds of lesion mask and the four artifact types are distributed almost identically over the three sets, so differences between validation and test results are unlikely to stem from a shift in composition. As external test sets we use Derm7pt [31], which provides a dermoscopic and a clinical image for each of its 1,011 cases (melanomas include *in situ* lesions and metastases), and the smartphone images of PAD-UFES-20 [32]. PH2 serves only to check the segmentation, because the copy available to us has no diagnoses. None of the external images is used for training or for choosing thresholds or models.

B. Models and training

The eight configurations differ only in the components listed below; backbone (ImageNet-pretrained DenseNet-121), input size (224×224), data, split and schedule are the same. **CNN**: backbone, global average pooling and a linear layer (black box). **PPN-std**: ProtoPNet with standard augmentation (random crop, horizontal flip). **PPN**: ProtoPNet with strong augmentation (random resized crop, scale $[0.4, 1]$; horizontal and vertical flips; $\pm 20\%$ brightness, contrast, saturation), used by all following models. **PPN-FA**: PPN with the IAIA-BL fine-annotation loss [16] on our lesion masks ($\lambda = 10^{-3}$), the mask-supervised setting of Correia *et al.* [12] with the same loss weight. **Ours**: lesion attraction and constrained projection; in the main runs the objective also contains the exploratory term $\mathcal{L}_{\text{art}}$, i.e. Eq. (1) with $\lambda_l = \lambda_a = 1$. **Ablations**: without $\mathcal{L}_{\text{les}}$ (Ours-noLes), without $\mathcal{L}_{\text{art}}$ (Ours-noArt, i.e. the two core components only), and with the standard projection (Ours-stdPush).

We train with Adam, a batch size of 64 and mixed precision. The first 5 epochs keep the backbone frozen and train the add-on layers and prototypes at a learning rate of 3×10^{-3}. The next 15 epochs train all layers (backbone 10^{-4}, weight decay 10^{-3}, learning rate divided by 10 every 5 epochs). The

TABLE II

THE ISIC 2019 SPLIT: IMAGES, DIAGNOSES, ACQUISITION SOURCES (IMAGES / MELANOMAS), PROVENANCE OF THE LESION MASKS AND PREVALENCE OF THE DETECTED ARTIFACTS; ALL PROPORTIONS ARE NEARLY IDENTICAL ACROSS THE THREE SETS. THRESHOLDS AND CHECKPOINTS ARE SELECTED ON VALIDATION. EXTERNAL TEST SETS: DERM7PT, 1,011 CASES WITH A DERMOSCOPIC AND A CLINICAL IMAGE EACH (252 MELANOMAS); PAD-UFES-20, 2,298 IMAGES (52 MELANOMAS); PH2, 200 IMAGES (SEGMENTATION ONLY).

	Train	Val.	Test	All
Images	20,552	2,459	2,320	25,331
Melanoma rate (%)	17.8	19.3	16.8	17.9
Diagnosis				
Melanoma	3,657	475	390	4,522
Nevus	10,523	1,208	1,144	12,875
Basal cell carc.	2,671	322	330	3,323
Actinic keratosis	693	80	94	867
Benign keratosis	2,095	278	251	2,624
Dermatofibroma	193	16	30	239
Vascular lesion	201	27	25	253
Squamous cell carc.	519	53	56	628
Source (images / melanomas)				
BCN20000	10,037/2,299	1,225/310	1,151/248	12,413/2,857
HAM10000	8,163/917	952/103	900/93	10,015/1,113
MSK	2,352/441	282/62	269/49	2,903/552
Lesion mask				
Expert (HAM10000)	8,163	952	900	10,015
Predicted (BCN, MSK)	12,389	1,507	1,420	15,316
Valid mask (%)	99.91	99.96	100.0	–
Artifact prevalence (%)				
Dark corners	25.5	25.7	25.1	–
Thin dark structures	24.2	25.7	23.4	–
Bright structures	13.8	14.4	13.3	–
Ink / patches	1.6	1.9	2.0	–

projection and 5 last-layer epochs follow; the CNN is trained for the same 20 epochs. Class weights are 0.608 and 2.81. For the CNN we keep the checkpoint with the best validation AUC, whereas prototype models are always evaluated after the projection, with the last-layer epoch chosen on validation (the prototypes are the same in all of these epochs). Each configuration is run with seeds 0, 1 and 2, which takes about 30 minutes per run on an NVIDIA T4.

C. Evaluation protocol

Classification. The threshold of each run is fixed on the ISIC validation set with the Youden index and then reused for every test set. AUC, balanced accuracy (BAcc), sensitivity and specificity are given as mean $\pm$ std over the seeds. For significance testing we average the probabilities of the three seeds (seed ensemble), compute 95% confidence intervals from 1,000 bootstrap resamples (resampling lesion groups on ISIC) and compare AUCs with the paired DeLong test [33], [34].

Attribution. For a prototype model we take the three most activated prototypes, upsample their similarity maps to 224×224 and keep the top 5% of pixels, i.e. the 95th-percentile region that ProtoPNet uses for visualization. For the CNN we use the class-activation map; with a global-average-pooling head it coincides with Grad-CAM [35] up to a positive factor [36]. From this hot region we compute the share that lies on the lesion and on artifacts, check whether its maximum falls on an artifact (pointing), and compute the *enrichment*, the artifact share divided by the artifact area. An enrichment above 1 means that artifacts get more attention than their size alone would explain.

Counterfactuals. Each test image is modified in four ways, always outside a 4-pixel margin around the lesion: we draw a ruler along one border, 12–29 curved hairs, a dark vignette, or 2–4 purple ink arcs. In the opposite direction, we erase the thin dark structures found by the detector (real hair and ruler marks, 544 images) with Telea inpainting [37], outside a 2-pixel margin. As a null control, a region of the same shape is shifted onto artifact-free skin and inpainted as well. A check after every modification confirms that each lesion pixel is bit-identical to the original. We then record the mean $|\Delta p|$ of the melanoma probability and the rate of flipped decisions. For the three prototypes most active on the original image we also measure how much the explanation moves: the Spearman correlation ρ of the top-1 map, the IoU of the hot regions and the *capture*, the part of the new hot region that lies on inserted pixels. Seed-averaged per-image values are compared with the Wilcoxon signed-rank test.

VI. RESULTS

Unless stated otherwise, comparisons in this section are between the five main models (CNN, PPN-std, PPN, PPN-FA, Ours); the ablations are discussed in Sections VI-F and VI-G. In all tables, bold marks the best of the four main prototype models.

A. Artifacts are associated with melanoma

In the training split, melanomas show dark corners and bright structures significantly more often than other lesions (Table III). This could simply reflect the source of the images, since dark corners appear in 47% of the BCN images but in only 5% of the HAM images, and BCN also has the highest melanoma rate. Stratifying by source does not remove the effect, however. The Mantel–Haenszel odds ratios remain clearly above 1, and for dark corners the odds ratio exceeds 1 in each source separately (BCN 1.41, HAM 3.10, MSK 2.24); for bright structures it does so in BCN and HAM (2.11, 1.92) but not in MSK (0.88). The strength of both associations differs between sources (Breslow–Day $p < 10^{-9}$), so the pooled odds ratios should be read as averages. For thin dark structures and ink we find no association.

B. Classification

On the ISIC test set, Ours reaches the highest AUC among the prototype models (0.865 ± 0.006, ensemble 0.884; Table IV). The difference to PPN, PPN-std and PPN-FA is significant (DeLong $p \leq 7.3 \times 10^{-4}$; $p \leq 9.0 \times 10^{-4}$ after Holm correction for the three comparisons), and the results vary less between seeds (std 0.006 compared with 0.040 for PPN). The ranking holds for every single run: each of the three Ours seeds is more accurate than each of the nine

TABLE III
ARTIFACT–MELANOMA ASSOCIATION (TRAINING SPLIT). OR: ODDS RATIO; MH: MANTEL–HAENSZEL OR STRATIFIED BY SOURCE.

Artifact	Prev.	$P(\mathrm{m}\|\mathrm{a})$	$P(\mathrm{m}\|\neg\mathrm{a})$	OR	MH OR [95% CI]
Dark corners	25.5%	.263	.149	2.03	1.555 [1.43, 1.69]
Bright structures	13.8%	.266	.164	1.85	1.751 [1.59, 1.92]
Thin dark struct.	24.2%	.174	.179	0.97	0.988 [0.90, 1.08]
Ink / patches	1.6%	.202	.178	1.18	1.114 [0.85, 1.46]
Any	48.8%	.210	.148	1.53	1.265 [1.17, 1.36]

baseline seeds (worst Ours seed 0.8592 vs. best baseline seed 0.8526), and the same holds on Derm7pt-dermoscopic (0.7405 vs. 0.7343). At the threshold fixed on validation, Ours trades some sensitivity for specificity (0.734/0.825 vs. 0.786/0.730 for PPN), and its balanced accuracy is the highest of the four main prototype models. The black-box CNN is still more accurate (ensemble 0.912 vs. 0.884), so interpretability keeps a price of about 3 AUC points; for PPN the gap is 4.6. The ranking does not depend on the last-layer re-optimization. With the five-epoch last layer of the training schedule (epoch chosen on validation), all prototype models score lower, but Ours remains the best: 0.846±0.007 vs. 0.797±0.027 (PPN-std), 0.770±0.027 (PPN) and 0.765±0.023 (PPN-FA), with the ablations at 0.787–0.841. Its ensemble is again significantly better than each prototype baseline (DeLong $p \leq 3.1 \times 10^{-4}$), and its AUC changes least (by 0.019 vs. 0.039–0.079 for the prototype baselines), which indicates that lesion-anchored prototypes keep their discriminative power after projection.

Externally, Ours has the highest AUC of the four main prototype models on Derm7pt-dermoscopic (0.749) and Derm7pt-clinical (0.657). On the dermoscopic images its ensemble is significantly better than PPN-FA ($p = 1.7 \times 10^{-5}$) and PPN-std ($p = 2.8 \times 10^{-6}$), also after Holm correction, but not than PPN ($p = 0.20$), whose seed variance is high; on the clinical images its advantage over PPN-std ($p = 0.023$) and PPN ($p = 0.046$) does not survive Holm correction ($p = 0.069, 0.092$), and over PPN-FA it is not significant ($p = 0.30$). On PAD-UFES-20, which contains only 52 melanomas, Ours is significantly better than PPN-std ($p = 0.001$) and does not differ significantly from PPN and PPN-FA ($p \geq 0.22$).

C. Where the models look

The black-box CNN is the model most drawn to artifacts (Table V, Fig. 2): in images that contain artifacts, 23.1% of its hot region lies on them, 1.5 times more than their area would suggest, and in 11.7% of the test images its maximum sits on an artifact. The prototype models look at artifacts less than their area would suggest, and Ours does so least: 4.8% of the hot region (PPN 6.6%, PPN-FA 10.2%), an artifact maximum in 1.3% of images (PPN 4.2%), and the largest lesion fraction (58.9%). All 20 prototypes of every Ours run are lesion patches, whereas PPN-FA projects on average 10.0 prototypes onto lesion and 2.0 onto artifact cells (Fig. 3). Most of the attention that falls on artifacts falls on dark corners (Table VI): 36.5% of the CNN hot region in images with

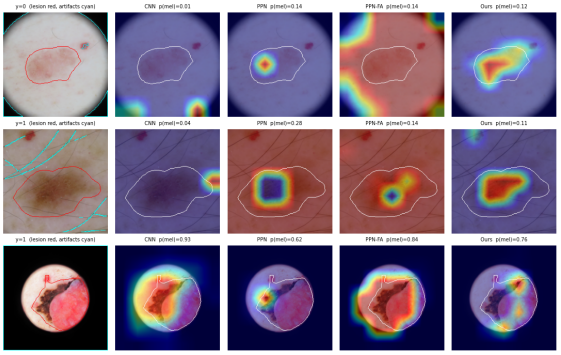


Fig. 2. ISIC test images with artifacts (left: lesion red, detected artifacts cyan). CNN: class-activation map; prototype models: map of the most activated prototype; p: melanoma probability (seed-0 models). Here the CNN and PPN-FA respond to dark corners and image borders, whereas the top prototype of Ours stays on the lesion.

dark corners, 9.8% for PPN, 15.8% for PPN-FA and 6.7% for Ours; thin dark and bright structures receive at most 2% from any model. All differences between Ours and each other main model in lesion share, artifact share, enrichment and pointing are significant (paired Wilcoxon test on per-image values averaged over seeds, $p \leq 1.3 \times 10^{-9}$). On the external Derm7pt-dermoscopic set, Ours again has the lowest artifact share, enrichment and pointing rate (1.5% vs. 4.4% for PPN and 15.3% for the CNN; $p \leq 8.6 \times 10^{-3}$), but there the three prototype baselines place significantly more of their hot region on the lesion (0.75–0.79 vs. 0.73; $p \leq 1.2 \times 10^{-4}$).

D. Counterfactual robustness

Table VII summarizes the interventions; in all 2,063–2,318 modified images per insertion type the lesion pixels were confirmed to be unchanged. Adding dark corners *increases* the melanoma probability of every model (mean signed change $+0.041$ CNN, $+0.048$ PPN, $+0.020$ Ours), which matches the sign of the association in Table III. The models have absorbed the shortcut, and Ours has absorbed it least. Ours roughly halves the flip rate of PPN for dark corners (10.9%→5.2%) and rulers (7.5%→3.6%), reduces it for ink (9.5%→5.6%) and hair (13.9%→9.2%), and flips fewer decisions than the CNN for all four insertions. These reductions are significant against every other main model for all four insertions (paired Wilcoxon test on per-image flips, $p \leq 3.9 \times 10^{-13}$ against the prototype baselines, $p \leq 0.022$ against the CNN), and so is the smaller per-image $|\Delta p|$ against PPN ($p < 10^{-39}$) and the CNN ($p \leq 0.018$). Removing real hair and ruler marks changes more decisions than the null control for every model (Ours 4.4% vs. 1.6%), so the measured sensitivity stems from the artifacts, not from inpainting.

Its explanations also move the least (Fig. 4). After adding dark corners its top-1 prototype map keeps $\rho = 0.92$ (PPN 0.68), and only 3.4% of its hot region moves onto the inserted pixels (PPN 6.1%, PPN-FA 11.4%; $p < 10^{-100}$). The exception is ink: the arcs lie just outside the lesion border, where lesion-anchored prototypes respond, so 1.8% of the hot

TABLE IV

CLASSIFICATION (MEAN ± STD OVER 3 SEEDS; THRESHOLD FROM ISIC VALIDATION). ENS.: SEED-ENSEMBLE AUC WITH 95% BOOTSTRAP CI; p: PAIRED DELONG TEST OF THE ENSEMBLE AGAINST OURS ON ISIC TEST.

	ISIC 2019 test (n = 2,320)						External AUC		
Model	AUC	BAcc	Sens.	Spec.	Ens. AUC [95% CI]	p	Derm7pt-derm	Derm7pt-clin	PAD-UFES-20
CNN (black box)	.902±.006	.814±.011	.808±.044	.820±.040	.912 [.884, .935]	3e-5	.799±.023	.704±.022	.721±.022
PPN-std	.835±.013	.752±.014	.769±.063	.734±.040	.861 [.827, .888]	6e-6	.677±.027	.624±.023	.587±.019
PPN	.825±.040	.758±.010	**.786±.068**	.730±.087	.866 [.830, .894]	7e-4	.688±.072	.618±.038	**.665±.016**
PPN-FA (IAIA-BL)	.844±.008	.744±.021	.689±.045	.800±.005	.866 [.830, .895]	4e-4	.696±.024	.640±.013	.630±.059
Ours	**.865±.006**	**.780±.014**	.734±.075	**.825±.046**	**.884 [.850, .910]**	–	**.749±.009**	**.657±.019**	.647±.108
Ours-noLes	.830±.015	.746±.030	.766±.091	.726±.032	.856 [.821, .885]	8e-8	.717±.016	.636±.033	.628±.047
Ours-noArt	.859±.011	.772±.003	.733±.056	.811±.054	.877 [.845, .904]	.085	.752±.014	.637±.030	.678±.018
Ours-stdPush	.863±.005	.785±.002	.823±.043	.747±.047	.884 [.852, .911]	.85	.747±.005	.670±.019	.658±.108

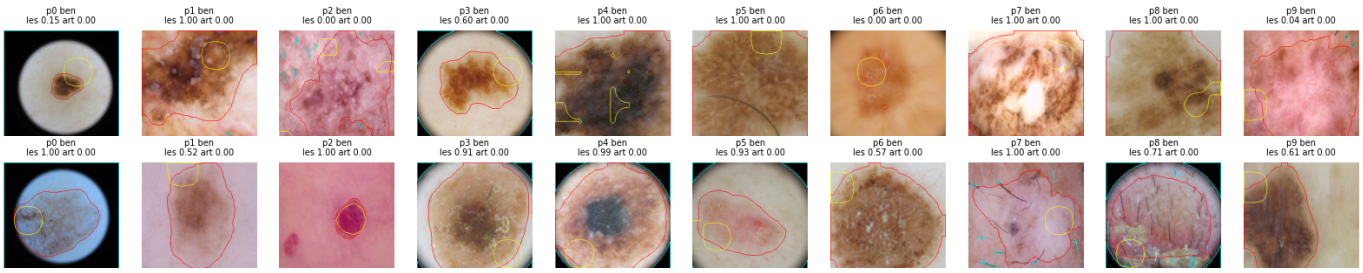


Fig. 3. Source patches of the ten benign-class prototypes of PPN (top) and Ours (bottom), seed 0 ("les"/"art": lesion and artifact fraction of the projected cell; red: lesion; cyan: artifacts; yellow: 95th-percentile region of the prototype on its own source image). Four PPN prototypes come from background skin (lesion fraction 0.00–0.15); all prototypes of Ours are lesion cells (lesion fraction ≥ 0.52), several of them at the lesion border.

TABLE V

WHERE THE MODELS LOOK (ISIC TEST, MEAN ± STD OVER SEEDS). ART.: HOT-REGION FRACTION ON ARTIFACTS IN IMAGES WITH ARTIFACTS; ENR.: ARTIFACT ENRICHMENT; POINT: MAXIMUM ON AN ARTIFACT; L-PR.: PROTOTYPES PROJECTED ONTO LESION CELLS (OF 20).

Model	Lesion↑	Art.↓	Enr.↓	Point↓	L-pr.
CNN	.466±.040	.231±.024	1.52±.46	.117±.007	–
PPN-std	.518±.036	.082±.049	0.54±.31	.050±.029	12.0
PPN	.562±.012	.066±.045	0.46±.04	.042±.040	14.3
PPN-FA	.530±.051	.102±.024	0.68±.05	.050±.006	10.0
Ours	**.589±.040**	**.048±.015**	**0.37±.13**	**.013±.005**	**20.0**
Ours-noLes	.565±.051	.106±.056	0.67±.11	.053±.029	20.0
Ours-noArt	.594±.028	.042±.008	0.40±.03	.018±.007	20.0
Ours-stdPush	.515±.063	.086±.052	0.58±.30	.034±.024	7.7

TABLE VI

ATTRIBUTION BY ARTIFACT TYPE (ISIC TEST; HOT-REGION SHARE ON THE GIVEN ARTIFACT IN IMAGES THAT CONTAIN IT) AND ON THE EXTERNAL DERM7PT-DERMOSCOPIC SET (MEAN OVER 3 SEEDS).

	ISIC test, by type				Derm7pt-derm		
Model	DC	TS	SP	INK	Lesion	Point	Enr.
CNN	.365	.016	.017	.029	.605	.153	.63
PPN-std	.123	**.010**	.015	.049	.749	.075	.25
PPN	.098	**.010**	.012	.036	**.789**	.044	.24
PPN-FA	.158	**.010**	**.010**	.056	.760	.077	.28
Ours	**.067**	.011	.016	**.032**	.734	**.015**	**.18**
Ours-noLes	.162	.011	.013	.044	.772	.057	.30
Ours-noArt	.056	.012	.020	.040	.713	.017	.23
Ours-stdPush	.129	.012	.019	.030	.633	.033	.35

region of Ours falls on ink vs. 1.0% for PPN, although Ours still flips fewer decisions.

E. Shortcuts in unedited images

The counterfactuals insert synthetic artifacts. As a complementary test without any editing, we ask whether the artifacts that are already present in the test images raise the melanoma score of *benign* lesions (Table VIII). We use a threshold-free measure, the probability A that a benign lesion with the artifact receives a higher score than a benign lesion without it ($A = 0.5$: no effect), and compare models on the same images with the DeLong test. On the ISIC test set, dark corners double the false-positive rate of PPN-FA (16%→33%) and raise that of Ours-noLes from 25% to 37%, whereas that of Ours rises only from 17% to 20%. The effect A of Ours (0.57) is significantly smaller than that of PPN, PPN-FA and Ours-noLes (0.61–0.66; $p \leq 1.6 \times 10^{-8}$) and similar to that of PPN-std and the CNN. The mask loss therefore does not remove the real dark-corner shortcut, whereas lesion attraction keeps it small. The same holds for bright structures, where Ours has the smallest effect of the prototype models (0.59 vs. 0.60–0.62; $p \leq 0.027$ against PPN-std and PPN-FA), and, most clearly, for dark border regions in the clinical Derm7pt photographs: 0.69 vs. 0.78–0.82 for the prototype baselines ($p \leq 1.4 \times 10^{-6}$), with false-positive rates of 24% vs. 12% without the artifact for Ours against 35–60% vs. 11–21% for the baselines. The CNN shows a similar or smaller effect, although it attends to dark corners most (Section VI-C).

TABLE VII

COUNTERFACTUAL INTERVENTIONS ON ISIC TEST (LESION PIXELS UNCHANGED; MEAN OVER 3 SEEDS). FLIP: DECISION CHANGES (%); $|\Delta p|$: MEAN ABSOLUTE CHANGE OF THE MELANOMA PROBABILITY; ρ: SPEARMAN CORRELATION OF THE TOP-1 PROTOTYPE MAP BEFORE/AFTER; CAP.: SHARE OF THE NEW HOT REGION ON INSERTED PIXELS (%). THE NULL CONTROL IS NOT RANKED.

	+Dark corners		+Ruler		+Hair		+Ink		Remove hair/ruler		Null control		Expl.: dark corners		Expl.: ruler	
Model	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	$\rho\uparrow$	Cap.$\downarrow$	$\rho\uparrow$	Cap.$\downarrow$
CNN	6.1	.049	5.6	.049	12.5	.104	9.7	.080	4.7	.036	1.0	.008	–	–	–	–
PPN-std	10.9	.060	8.3	.041	14.9	.092	9.9	.058	8.8	.036	4.3	.013	.635	7.1	.773	1.5
PPN	10.9	.067	7.5	.045	13.9	.096	9.5	.062	7.4	.045	3.6	.019	.680	6.1	.804	1.1
PPN-FA	10.6	.049	6.5	.028	15.1	.084	9.1	.046	7.2	**.030**	3.2	.010	.714	11.4	.874	1.1
Ours	**5.2**	**.035**	**3.6**	**.025**	**9.2**	**.081**	**5.6**	**.045**	**4.4**	.032	1.6	.011	**.918**	**3.4**	**.952**	**0.7**
Ours-noLes	10.2	.053	8.7	.040	14.3	.083	10.1	.049	8.3	.042	3.9	.016	.705	8.6	.827	1.7
Ours-noArt	5.4	.036	3.5	.023	8.7	.061	6.1	.041	4.4	.027	1.0	.011	.884	3.4	.949	1.0
Ours-stdPush	4.5	.034	4.1	.028	9.6	.077	5.8	.045	4.5	.033	1.9	.011	.923	7.5	.944	1.4

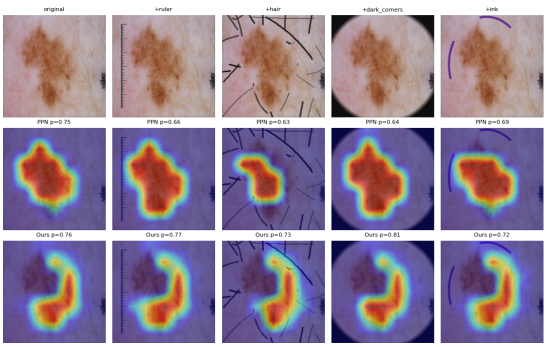


Fig. 4. Counterfactual example with unchanged lesion pixels (the evaluation notebook draws the first two test images that receive a ruler; this is the second, the first is in the repository). Rows 2–3: map of the prototype most active on the original image and melanoma probability p for PPN and Ours. The map of Ours keeps its place under all four insertions, whereas that of PPN shifts when hair is added.

TABLE VIII

EFFECT OF REAL, UNEDITED ARTIFACTS ON BENIGN TEST LESIONS. A: PROBABILITY THAT A BENIGN LESION WITH THE ARTIFACT RECEIVES A HIGHER MELANOMA SCORE THAN ONE WITHOUT IT (SEED ENSEMBLES; 0.5 = NO EFFECT). FPR: FALSE-POSITIVE RATE (%) ON BENIGN LESIONS WITH/WITHOUT THE ARTIFACT AT THE VALIDATION THRESHOLD (MEAN OVER SEEDS). BENIGN IMAGES WITH/WITHOUT: 447/1,483 (ISIC, DC), 251/1,679 (ISIC, SP), 515/244 (DERM7PT-CLINICAL, DC).

	ISIC, dark corners		ISIC, bright struct.		Derm7pt-clin., DC	
Model	$A\downarrow$	FPR	$A\downarrow$	FPR	$A\downarrow$	FPR
CNN	.57	22/17	.58	19/18	.59	18/9
PPN-std	.58	31/25	.61	32/26	.82	60/21
PPN	.61	30/26	.60	33/26	.78	35/11
PPN-FA	.66	33/16	.62	26/19	.80	46/15
Ours	**.57**	**20/17**	**.59**	**19/17**	**.69**	**24/12**
Ours-noLes	.66	37/25	.63	37/26	.77	50/24
Ours-noArt	.58	20/19	.62	25/18	.75	37/12
Ours-stdPush	.57	28/25	.59	29/25	.65	33/19

F. Ablation of the two core components

Lesion attraction is the term behind the accuracy gain. Without it (Ours-noLes) the test AUC falls to 0.830 (DeLong $p = 8 \times 10^{-8}$), the artifact share of the hot region more than doubles (4.8%→10.6%), and the flip rates for dark corners, rulers and ink roughly double (Tables IV, V and VII). Ours-noLes still uses the constrained projection, so all of its prototypes are lesion patches; the comparison therefore shows that a correct projection alone is not enough, the attention during training has to be anchored as well.

The constrained projection leaves the accuracy unchanged (Ours-stdPush, $p = 0.85$) but determines what the prototypes *are*: with the standard projection only 7.7 of 20 prototypes lie on the lesion, the artifact share of the hot region rises from 4.8% to 8.6%, and the share captured by added dark corners from 3.4% to 7.5% ($p < 10^{-200}$). Accuracy alone would have hidden this difference, which is why we evaluate explanations directly.

G. Artifact suppression: a largely negative result

We also explored a term that penalizes attention on artifacts directly. With $A_{ij} = \sum_{h,w} \alpha_{ij} M_i^A$, the attention mass of prototype j on the artifacts of image i,

$$\mathcal{L}_{\text{art}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{j=1}^{m} \omega_{ij} A_{ij}. \tag{3}$$

Removing it (Ours-noArt) does not change the accuracy significantly, on ISIC ($p = 0.085$) or on any external set ($p \geq 0.058$); several counterfactual measures are even significantly better without it (e.g., $|\Delta p|$ 0.061 vs. 0.081 for added hair, Wilcoxon $p < 10^{-39}$), and the attribution changes little (artifact share 4.2% vs. 4.8%, $p = 0.40$; artifact pointing 1.8% vs. 1.3%, $p = 2.2 \times 10^{-4}$). The two core components alone therefore carry the explanation gains: against every prototype baseline, Ours-noArt places more of its hot region on the lesion and less on artifacts, and flips fewer decisions under all four insertions (Wilcoxon $p \leq 1.8 \times 10^{-10}$). Their accuracy gain is significant against PPN-std (DeLong $p = 0.005$), whereas the smaller gains over PPN and PPN-FA (about $+0.01$ in ensemble AUC; $p = 0.040$ and 0.055) do not survive Holm correction ($p = 0.079$). The clearest benefit of $\mathcal{L}_{\text{art}}$ is a weaker response to real bright structures and to dark border regions in the clinical images (A 0.59 vs. 0.62 and 0.69 vs. 0.75; $p \leq 3.3 \times 10^{-5}$, Table VIII). Its effect is otherwise small because $\mathcal{L}_{\text{les}}$ already acts on a normalized attention, which

lowers the mass on artifacts automatically, and the constrained projection removes the remaining route by which an artifact could become a prototype. We therefore do not count $\mathcal{L}_{\mathrm{art}}$ as part of the method; Ours in the tables differs from the two-component model only by this term.

H. Robustness to mask quality

Predicted masks could bias the results if a model looked "on the lesion" only by the measure of an imperfect mask, or if the gains appeared only where masks were predicted. We therefore split the ISIC test set by mask provenance (900 HAM10000 images with expert masks, 1,420 BCN20000 and MSK images with predicted masks) and repeat the attribution with lesion masks eroded or dilated by 5 pixels (Table IX).

Expert masks alone. On the HAM10000 test images, whose masks were drawn by experts and not by our network, Ours places 53.4% of its hot region on the lesion, against 41.6–44.6% for the prototype baselines and 44.4% for the CNN; its advantage in lesion focus is thus *larger* on expert masks (+8.8 points over the best baseline) than on the whole test set (+2.7 points), so it is not produced by errors of the predicted masks. HAM10000 images contain few and small artifacts, so all artifact scores are low there, and those of Ours are the lowest. On the images with predicted masks, 5.6% of the hot region of Ours lies on artifacts (PPN 7.9%, PPN-FA 12.3%, CNN 27.7%), although PPN places a slightly larger fraction on the lesion (63.5% vs. 62.4%, within one standard deviation).

Perturbed masks. Eroding or dilating every lesion mask by 5 pixels (at 256×256) shifts the lesion fractions of all models in the same direction; the order of the five main models stays the same (Ours $>$ PPN $>$ PPN-FA $>$ PPN-std $>$ CNN), with a margin of Ours over PPN of 0.9 points under erosion and 4.5 under dilation. Artifact share, pointing and enrichment change by at most 0.011.

Classification per source. Ours is the most accurate prototype model in every source, both where its training masks were drawn by experts (HAM10000: 0.906 vs. 0.896 for PPN-FA) and where they were predicted (BCN20000: 0.852 vs. 0.834; MSK: 0.761 vs. 0.711 for PPN-std).

Counterfactuals per provenance. The counterfactual protocol uses the lesion mask to decide where artifacts may be drawn; if a predicted mask were too small, inserted artifacts could touch the true lesion. This is not what drives the results. On the expert-mask subset, where the protected region is exact, Ours flips the fewest decisions of all models (dark corners 4.4% vs. 7.1% for the CNN and 10.0–10.3% for the prototype baselines; rulers 2.3% vs. 3.9% and 4.7–6.1%). The predicted-mask subset gives the same picture (dark corners 5.7% vs. 10.7–11.5%; rulers 4.4% vs. 6.7% and 7.7–9.7%), except that the CNN flips slightly fewer decisions for dark corners (5.5%). The mean $|\Delta p|$ of Ours is also the smallest in both subsets (0.035–0.036 for dark corners vs. 0.043–0.068).

VII. Discussion and Limitations

The two components play different roles: lesion attraction changes what the network learns and makes it both more accurate and less dependent on artifacts, whereas the constrained projection changes what it shows to the user. Neither role is visible in the AUC alone.

Dependence on masks. Lesion masks are the main practical requirement, but only during training; deployment needs nothing but the image. The masks derive from public expert segmentations of a single source, so a new dermoscopic dataset needs no new annotation, and both components use them only through coarse 7×7 cell fractions, which tolerate boundary errors; Section VI-H confirms the conclusions on expert masks alone and under ± 5-pixel mask errors.

Limitations. (i) A gap of about 3 AUC points to the black-box CNN remains; Correia *et al.* [12] found prototype models on par with CNNs, but on ISIC validation data and after choosing the best of many configurations. (ii) For BCN20000 and MSK no expert masks exist, so the predicted masks of these sources can only be validated indirectly (held-out HAM10000 images, PH2, and the subset analysis). The bright-structure detector also responds to light hairs, and other artifacts (e.g., gel borders) are not modeled. (iii) Inserted artifacts are synthetic; we mitigated this with the removal of real artifacts, a null control and the test on unedited images, whose artifact effects may, however, be partly confounded by lesion appearance. (iv) The 7×7 latent grid limits the spatial resolution of the explanations, and class-activation and prototype maps are different kinds of explanation, so attribution comparisons between the CNN and the prototype models are only indicative. (v) External accuracy gains are significant against two of the three prototype baselines on Derm7pt-dermoscopic, against none after Holm correction on Derm7pt-clinical, and against one on the small PAD-UFES-20 melanoma set; a single backbone and a single prototype architecture were studied, as required for a controlled comparison, although both components apply to any prototype network with similarity maps and a projection step.

VIII. Conclusion

Standard and prototype classifiers trained on ISIC 2019 do exploit artifact shortcuts, and a prototype network can be anchored to the lesion without any new annotation. Two components suffice: lesion attraction on the prototype attention and a projection restricted to artifact-free lesion patches; an additional artifact-suppression term brought no significant gain in accuracy or counterfactual robustness. In our same-backbone, leakage-controlled comparison the lesion-anchored model was the most accurate prototype model, relied least on artifacts, changed its decisions and explanations least when artifacts were inserted or removed around an unchanged lesion, and was the prototype model least affected by real artifacts in unedited images, where the mask loss of the closest prior approach left a strong dark-corner shortcut. The counterfactual protocol does not depend on the architecture and can be used to check other interpretable classifiers as well.

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TABLE IX

ROBUSTNESS TO MASK QUALITY (ISIC TEST, MEAN ± STD OVER 3 SEEDS). LEFT: ATTRIBUTION MEASURED SEPARATELY ON IMAGES WITH EXPERT MASKS (HAM10000, $n = 900$) AND WITH PREDICTED MASKS (BCN20000 AND MSK, $n = 1{,}420$); LESION, ART. AND POINT AS IN TABLE V. MIDDLE: LESION FRACTION OF THE HOT REGION WHEN EVERY LESION MASK IS ERODED (-5 PX) OR DILATED ($+5$ PX). RIGHT: TEST AUC PER ACQUISITION SOURCE (THE CLASSIFIER RECEIVES NO MASK AT TEST TIME; n/MELANOMAS: BCN 1,151/248, HAM 900/93, MSK 269/49).

	Expert masks (HAM10000)			Predicted masks (BCN, MSK)			Lesion, perturbed mask		AUC by source		
Model	Lesion↑	Art.↓	Point↓	Lesion↑	Art.↓	Point↓	-5 px	$+5$ px	BCN	HAM	MSK
CNN	.444±.044	.043±.014	.015±.006	.480±.038	.277±.026	.182±.010	.434±.038	.496±.040	.883±.012	.934±.003	.861±.002
PPN-std	.416±.085	.016±.005	.004±.002	.582±.046	.098±.059	.079±.047	.480±.036	.552±.035	.825±.018	.880±.016	.711±.022
PPN	.446±.053	.014±.001	.005±.005	**.635±.015**	.079±.056	.065±.062	.525±.013	.595±.012	.819±.051	.860±.034	.692±.016
PPN-FA	.423±.045	.019±.003	.004±.002	.599±.060	.123±.030	.080±.011	.491±.055	.567±.047	.834±.009	.896±.004	.699±.014
Ours	**.534±.062**	**.013±.005**	**.002±.001**	.624±.026	**.056±.018**	**.021±.008**	**.534±.039**	**.640±.041**	**.852±.008**	**.906±.006**	**.761±.018**

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